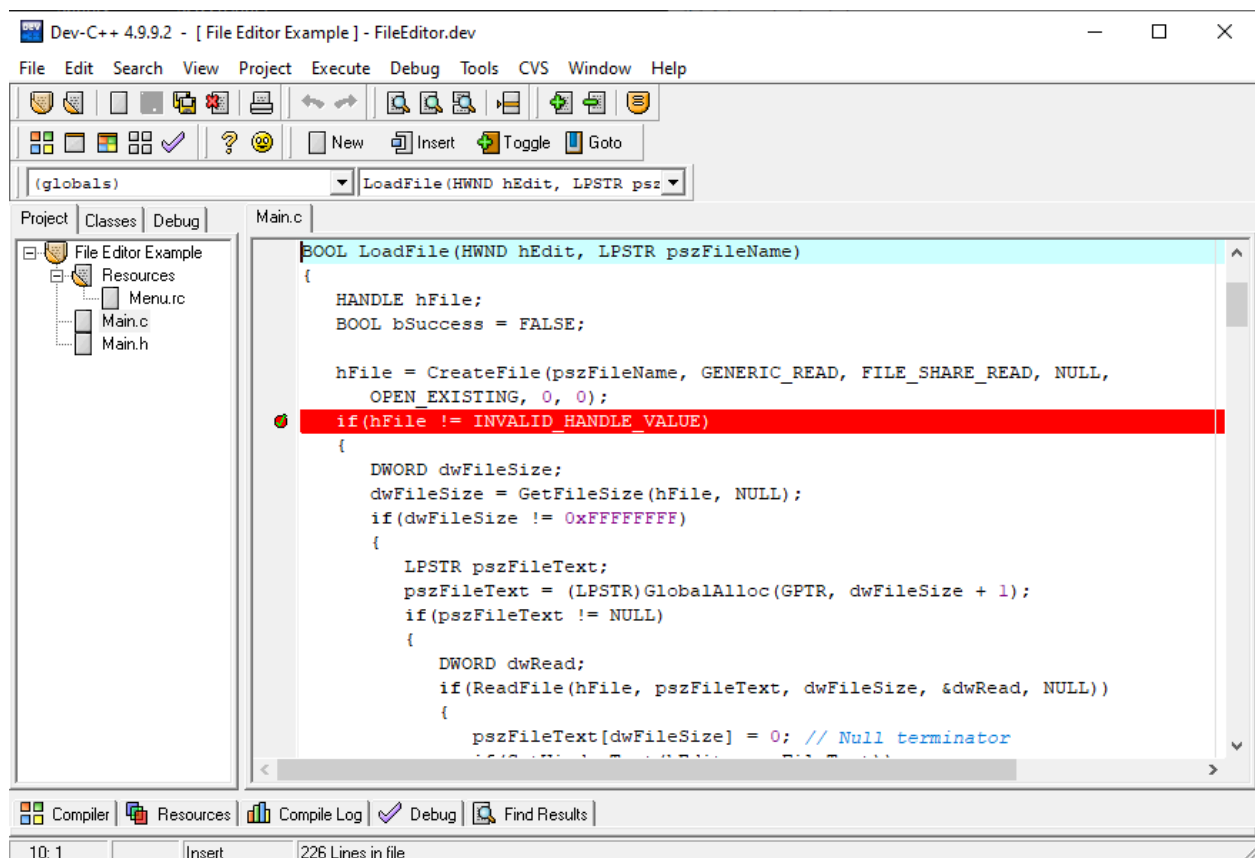


## DEV C++

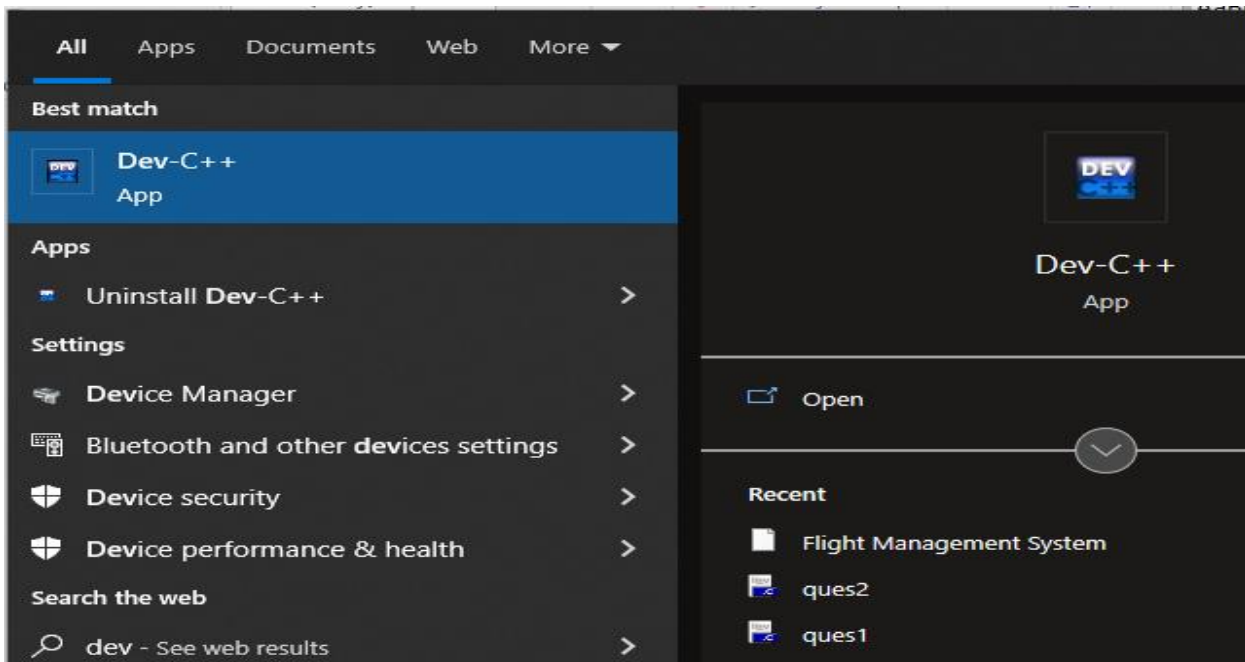
### Installation and Getting Started under Windows OS

Dev-C++ is a full-featured C and C++ Integrated Development Environment (IDE) for Windows platforms. Millions of developers, students and researchers use Dev-C++ since the first version was released in 1998. It has been featured in dozens of C++ and scientific books and remains one of the favorite learning tool among universities & schools worldwide.

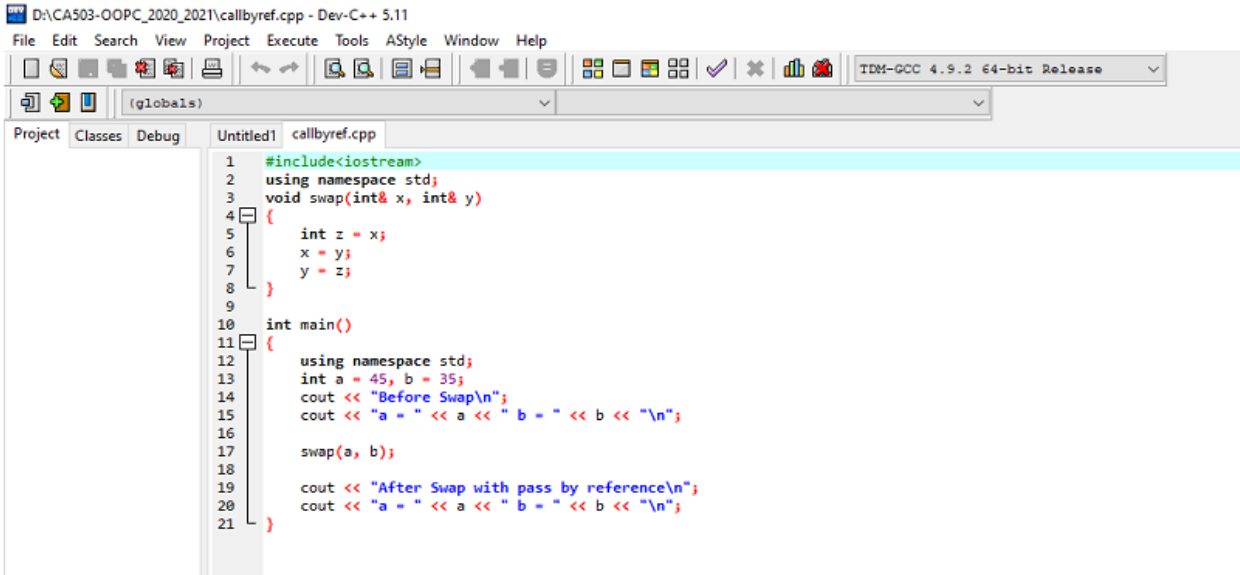
- 1 • Go to <https://www.bloodshed.net/>
- 2 • Click on [Download Original Dev-c++ for Windows](#)
- 3 • Click on [install Dev C++ for the first time](#)
- 4



## Tool Based Manual || BCA || SEM II || CA129-Fundamentals of Object Oriented Programming



Select Dev-c++ from start menu,it open Dev-c++ editor

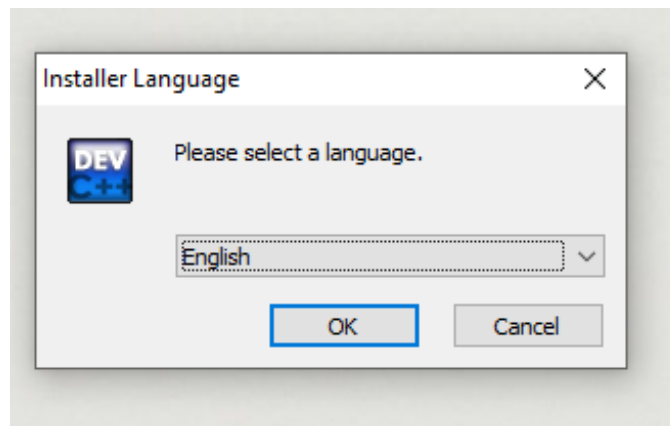


Write your code in Editor, save it and Select Execute->Compile and Execute->Run (F10)

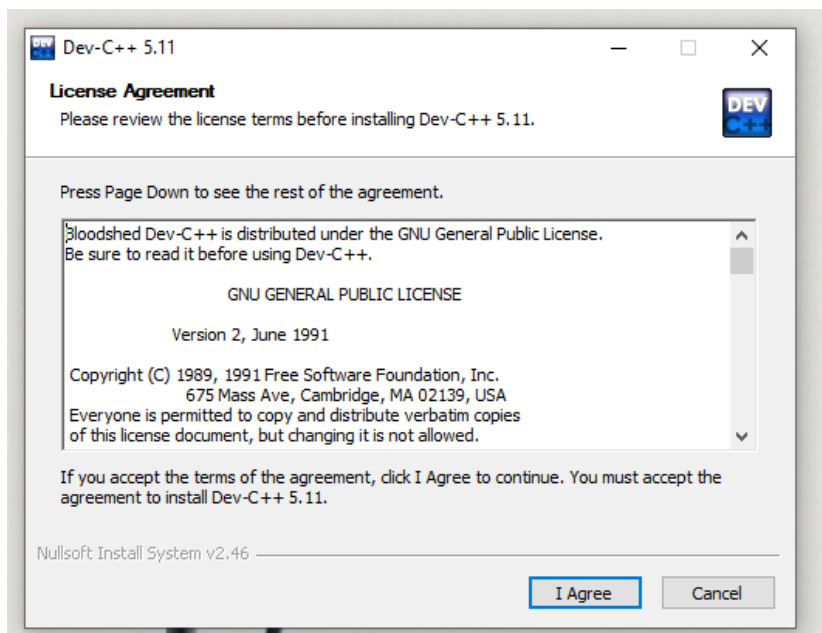
## Installation and Getting Started under Windows OS

Follow the steps mentioned in following figure to install Dev c++ on Windows Operating System.

1. Go to <https://sourceforge.net/projects/orwelldvcpp/>
2. Click on Download and wait for five seconds.
3. Double click on Dev-Cpp 5.11 TDM-GCC 4.9.2 Setup file

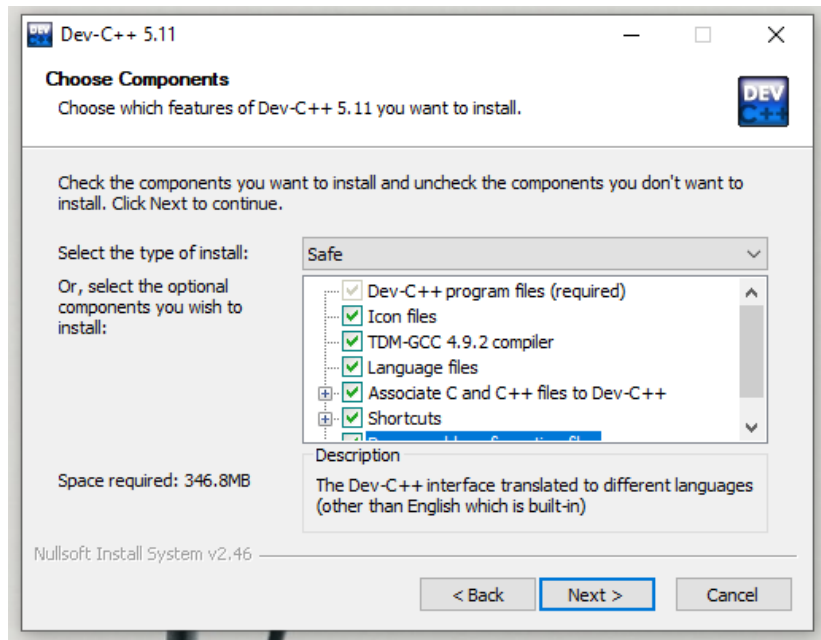


4. Click On Ok.

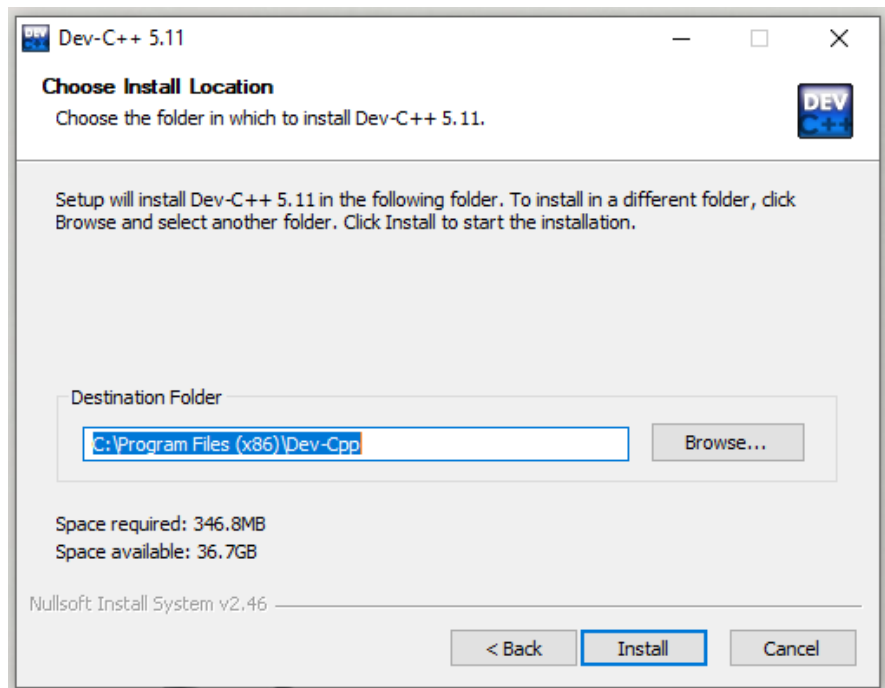


## Tool Based Manual || BCA || SEM IV || CA221 : Data Structure and Algorithms

5. Click On I Agree.



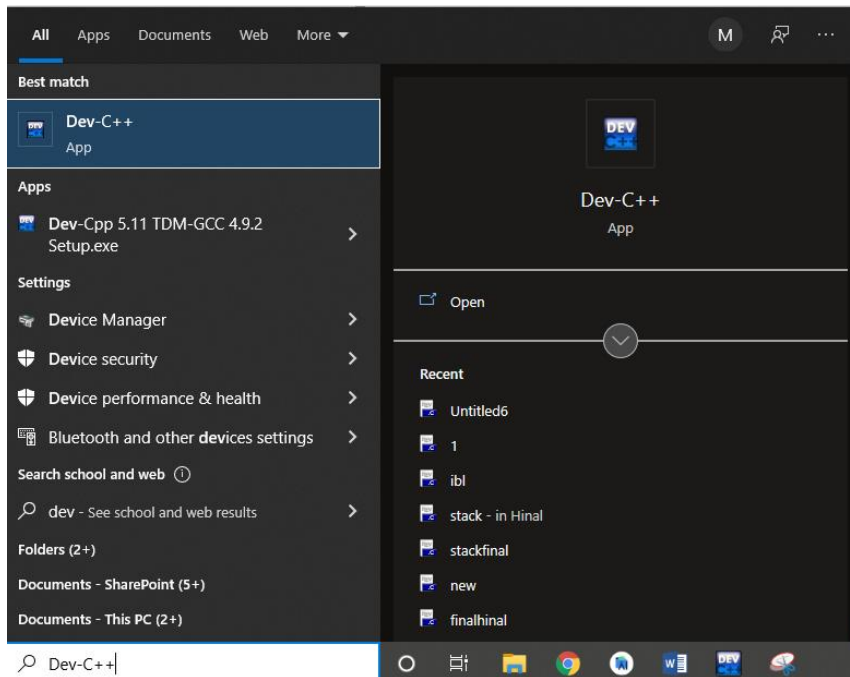
6. Click on Next.



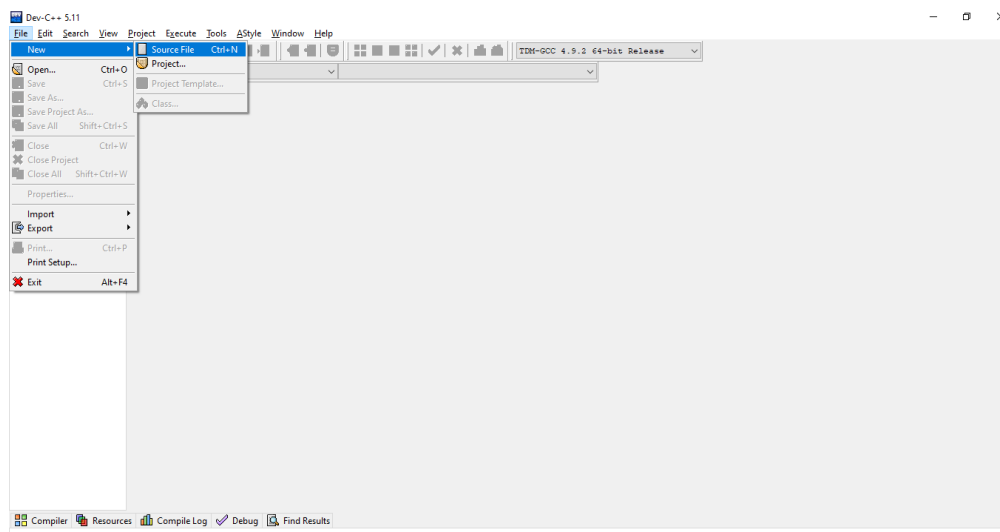


## Tool Based Manual || BCA || SEM IV || CA221 : Data Structure and Algorithms

9. Click on Next, Next Okay.

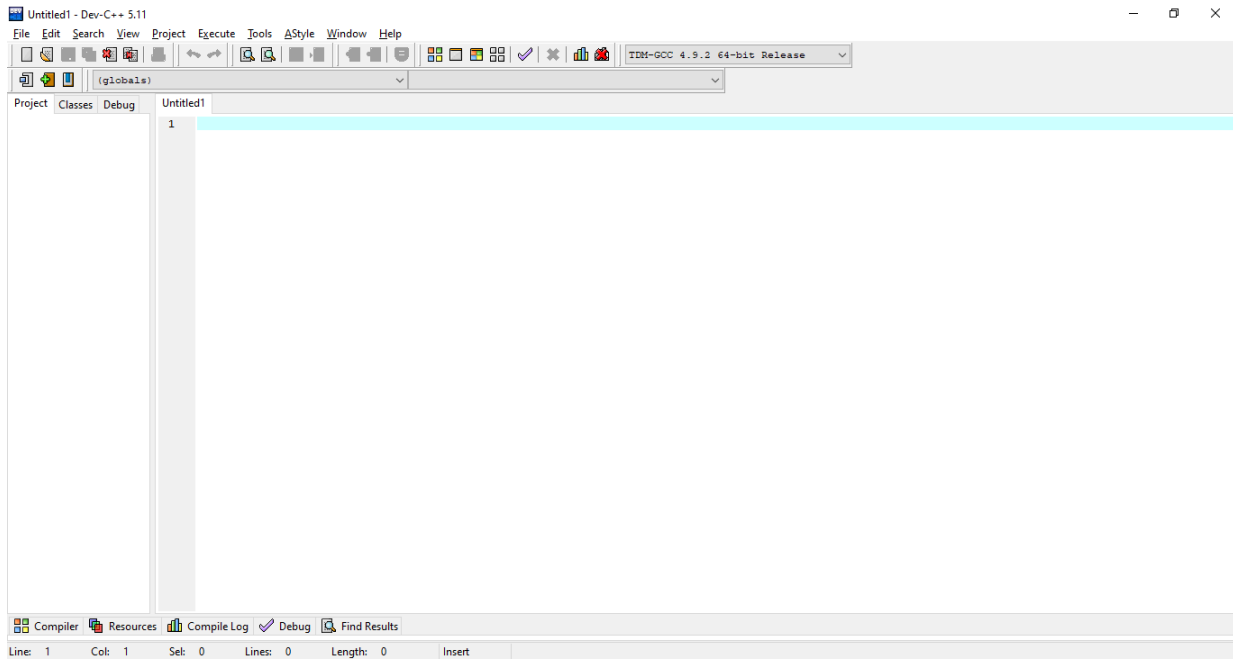


10. Open Dev C++.

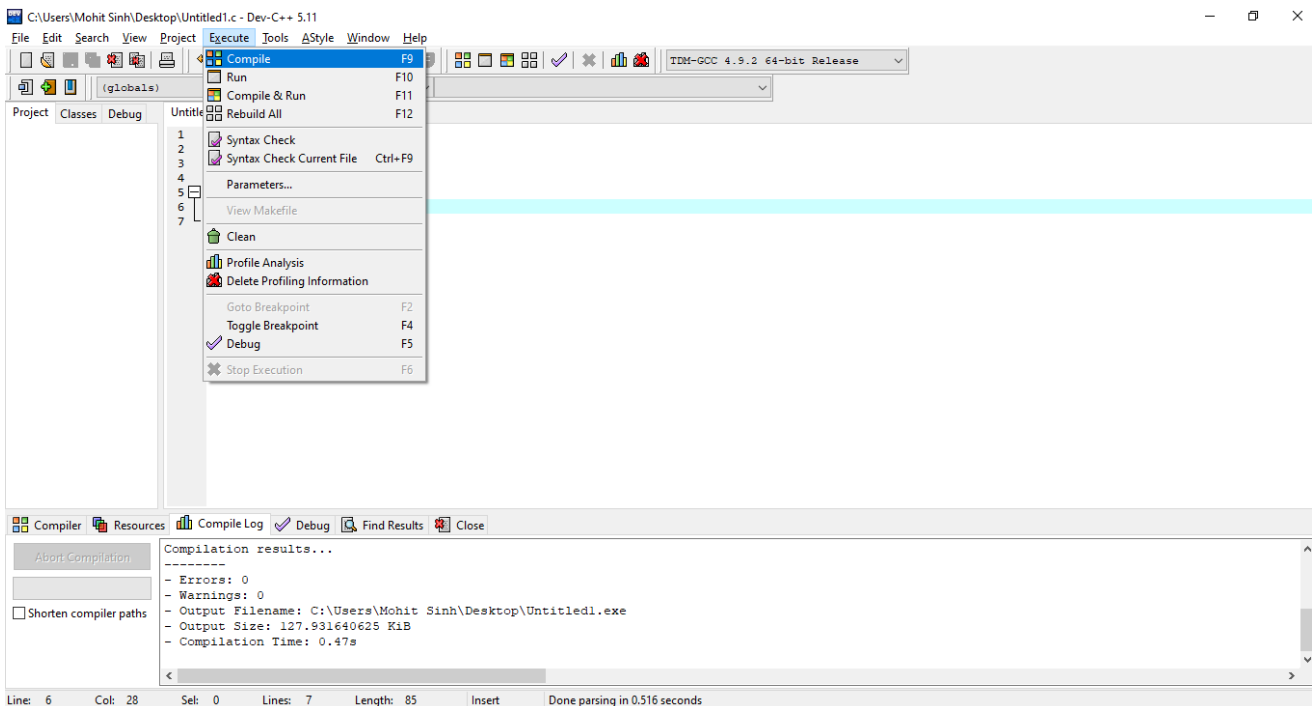


## Tool Based Manual || BCA || SEM IV || CA221 : Data Structure and Algorithms

### 11. IDE of Dev C++.



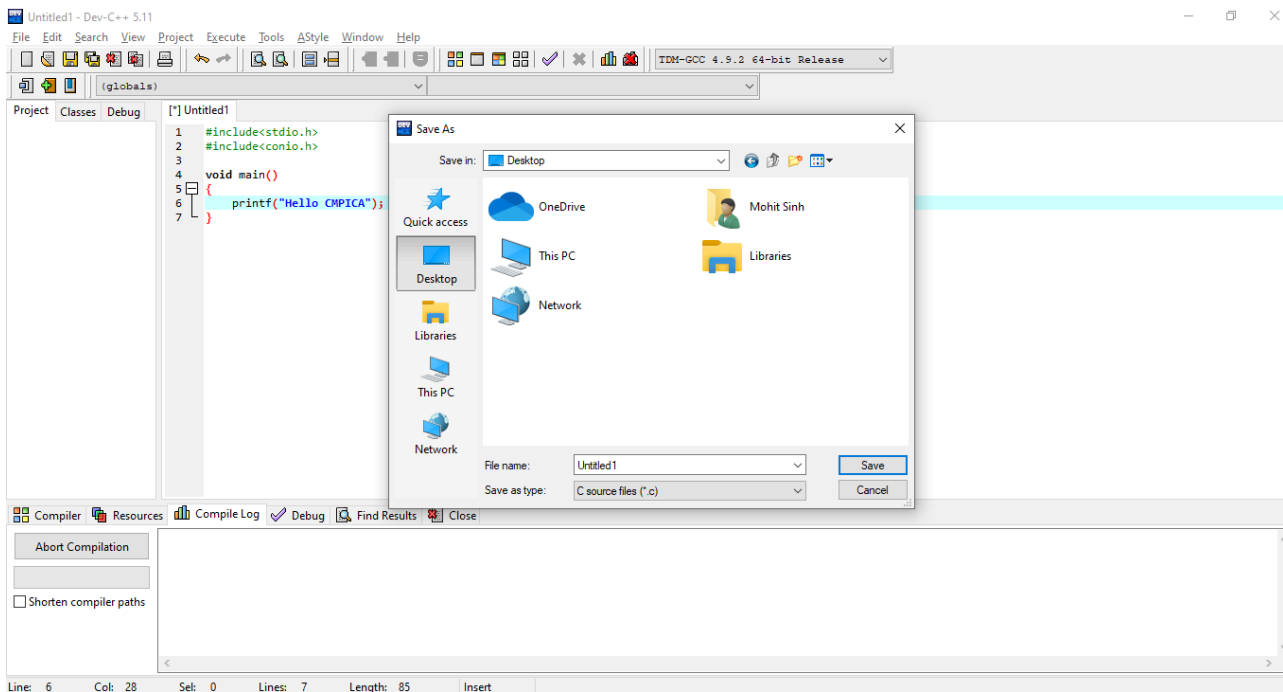
### 12. New file for c programming.



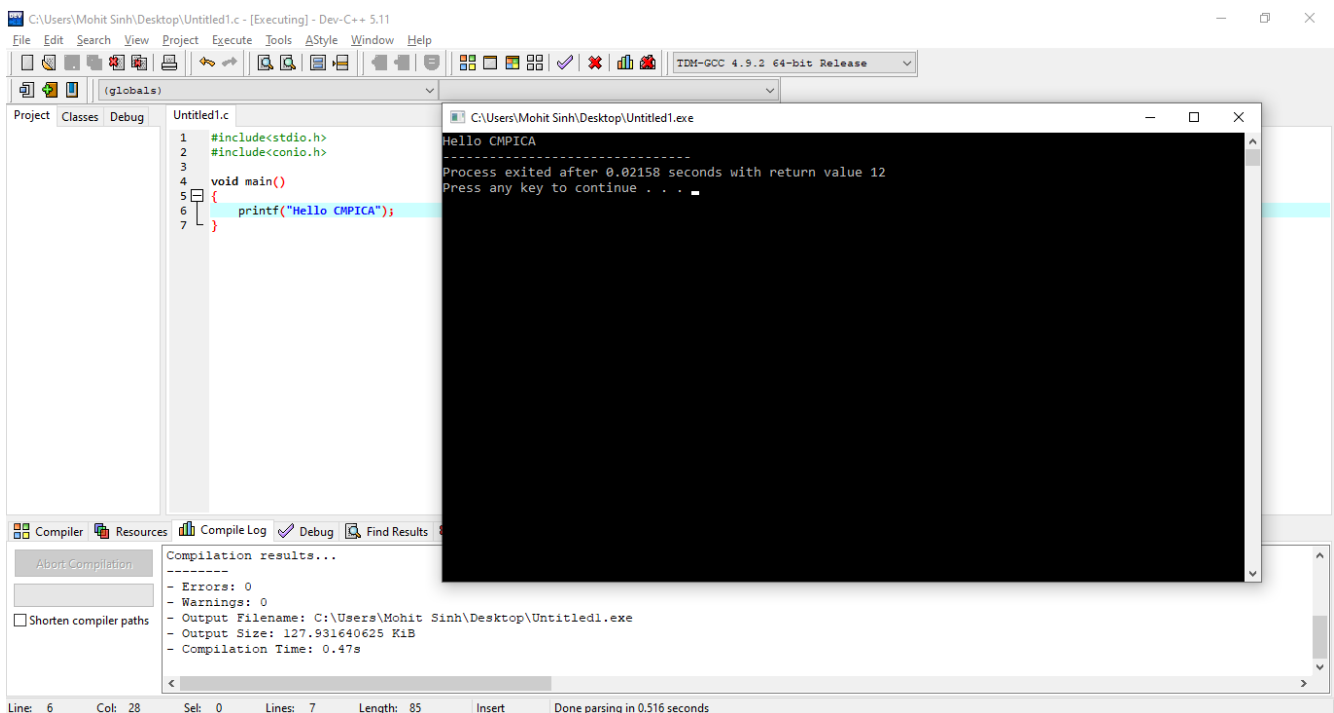


## Tool Based Manual || BCA || SEM IV || CA221 : Data Structure and Algorithms

13. From the execute tab of menu bar you have to select compile option.



14. To run the program, save the file with appropriate name and extension.



The black screen window is the console window that display the output of the program.



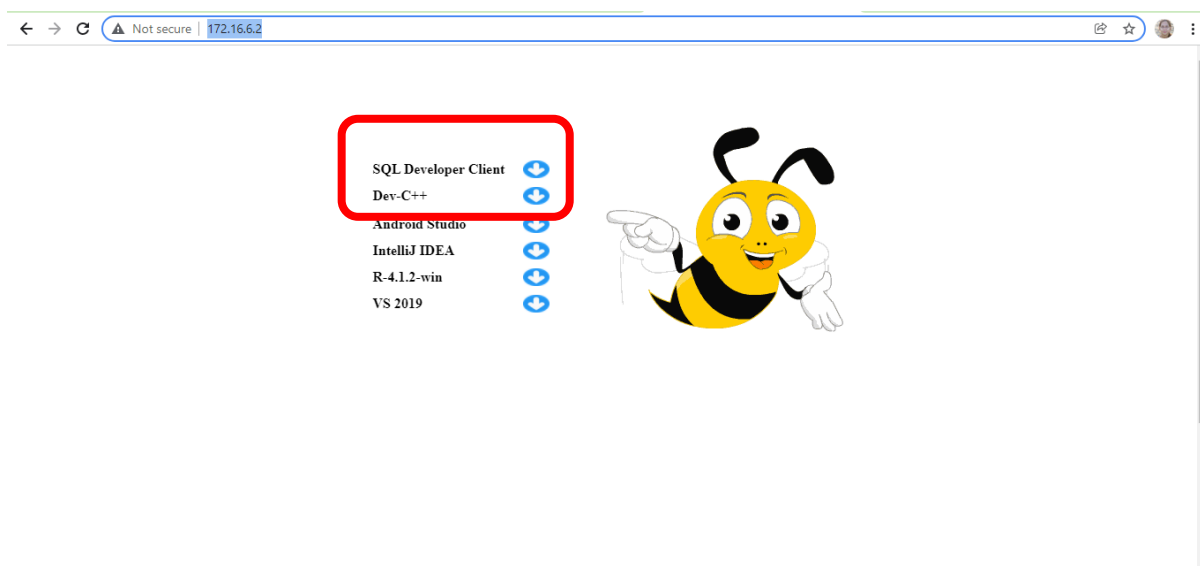
## Tool Based Manual || BCA || SEM IV || CA222 : Advanced Database System

### Objectives

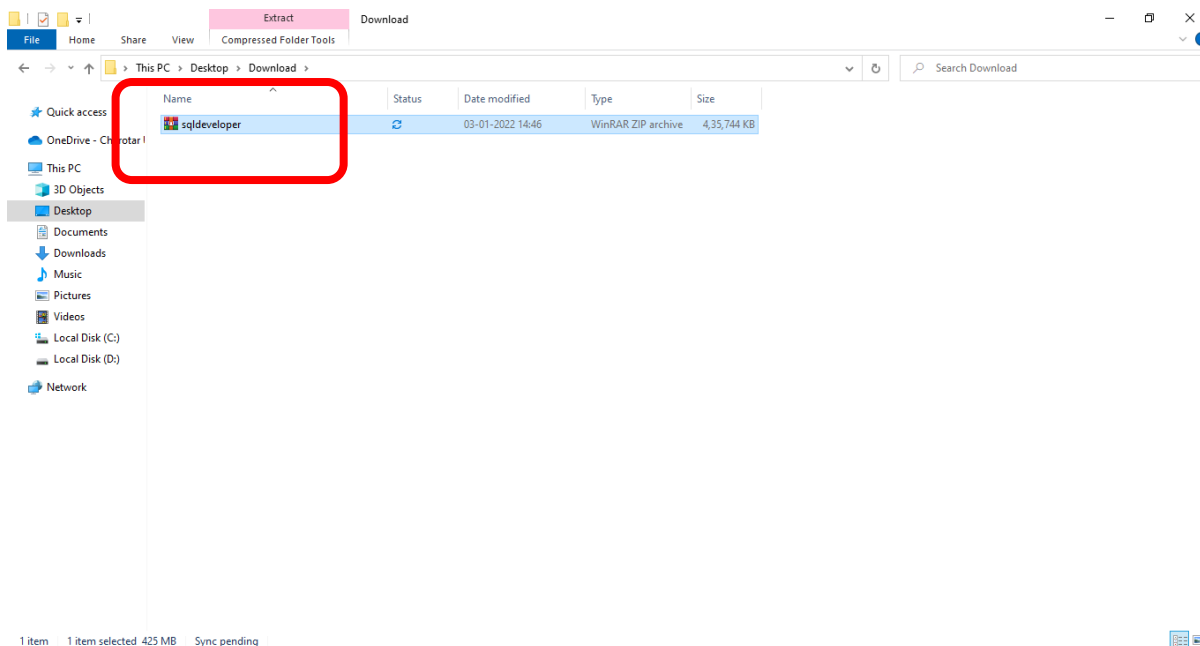
- To make students acquaint with installation of Database Client Software.

**Step: 1** Go to <http://172.16.6.2/>

Download SQL Developer Client

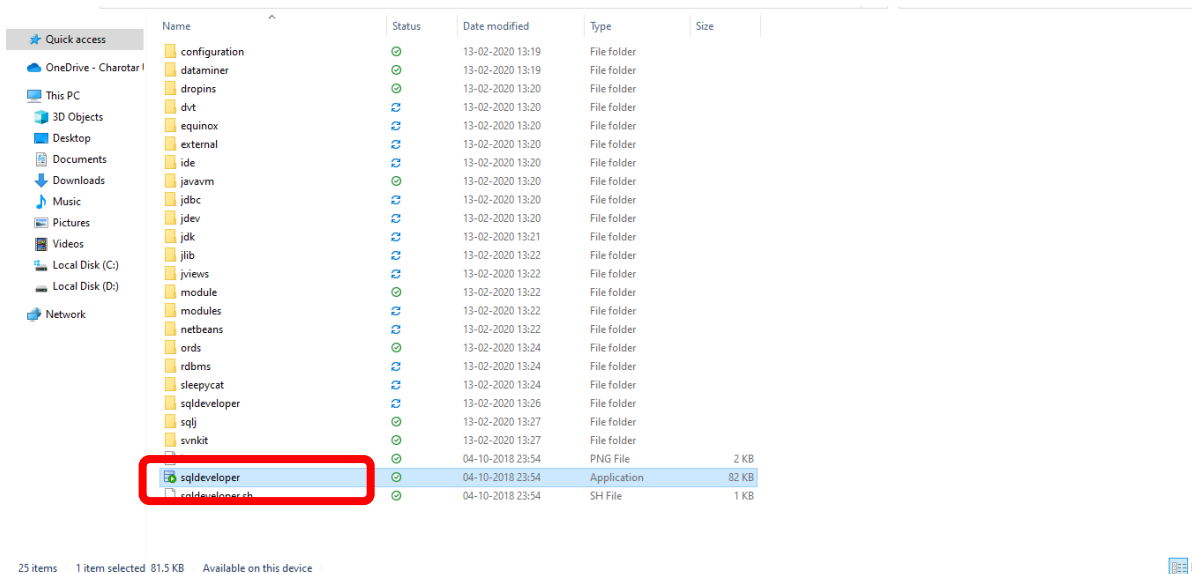


This will download the zip with name **sqldeveloper**. Unzip it.

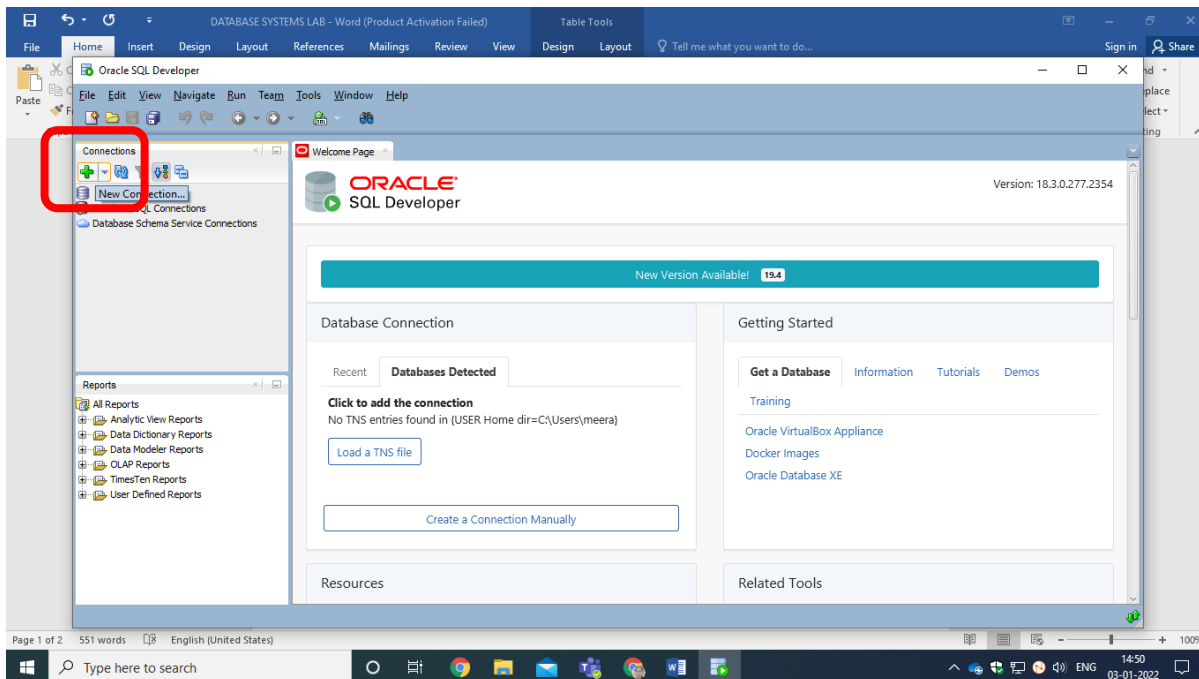


**Step: 2** Double click on **sqldeveloper** Application

## Tool Based Manual || BCA || SEM IV || CA222 : Advanced Database System

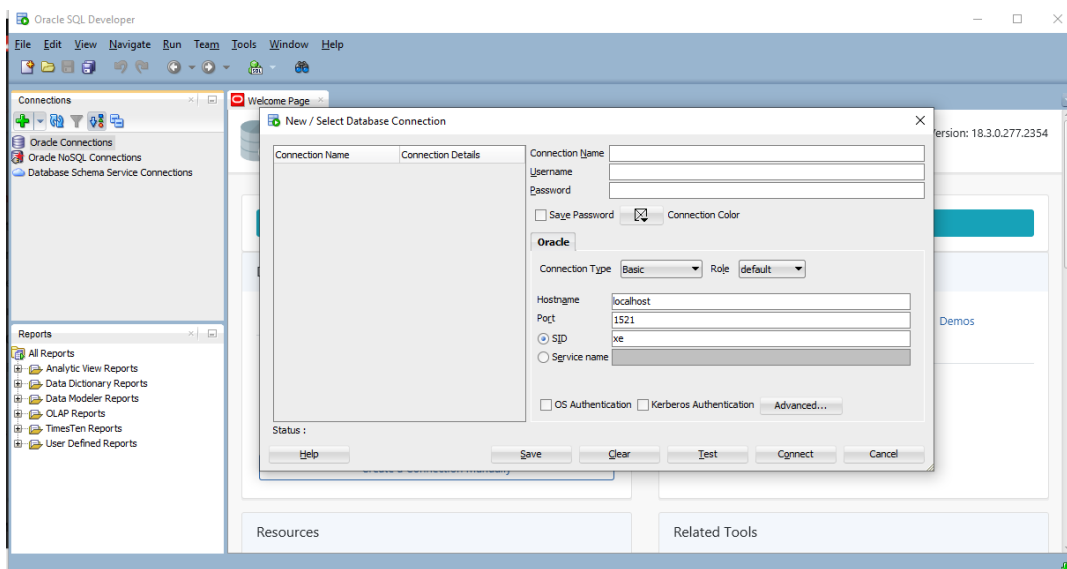


It will open the Oracle SQL Developer.



**Step: 3** Click on  Sign, the following screen will appear.

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**Step: 4** Give Proper Connection Name for example CMPICA

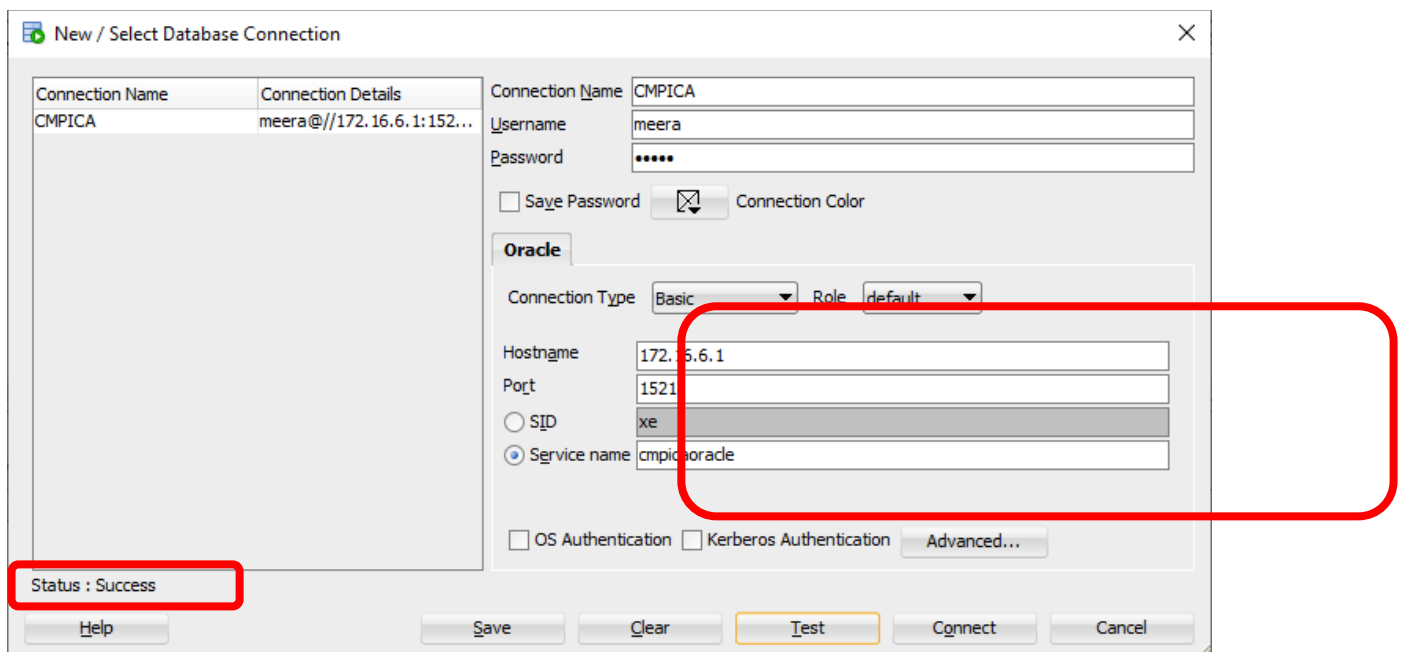
Enter Username and Password provided to you in respective fields.

In Hostname write **172.16.6.1**

Click service name radio button and enter **cmpicaoracle**

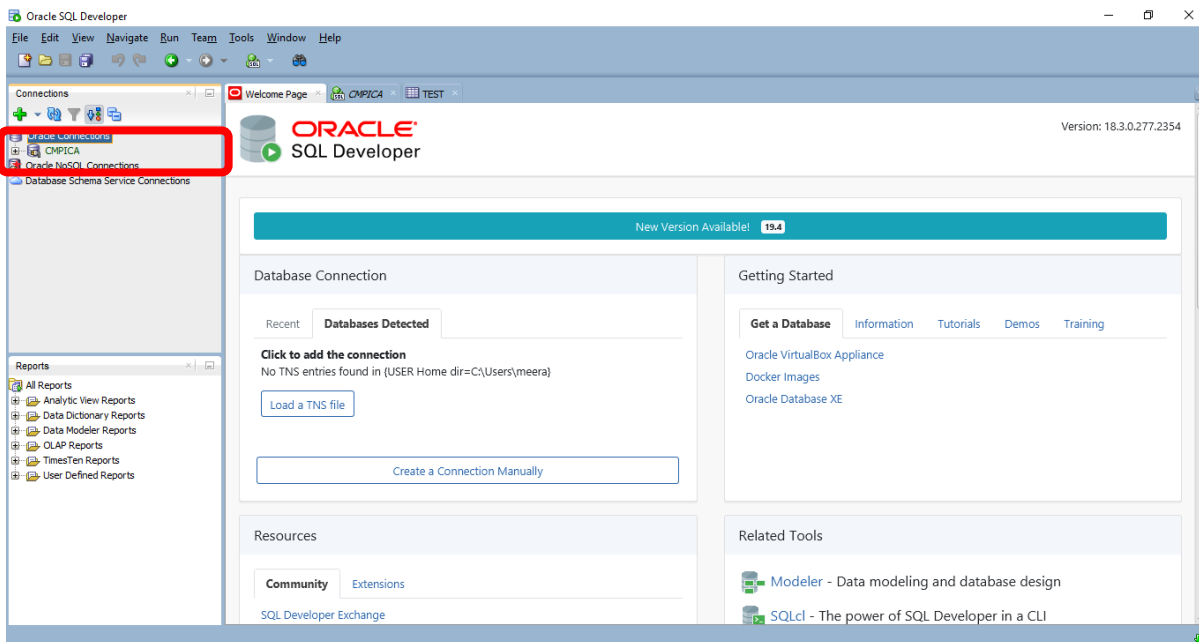
KEEP REST OF THE CONFIGURATION AS IT IS. And Click on Test button. The **success** should get displayed.

Now click on Connect Button.

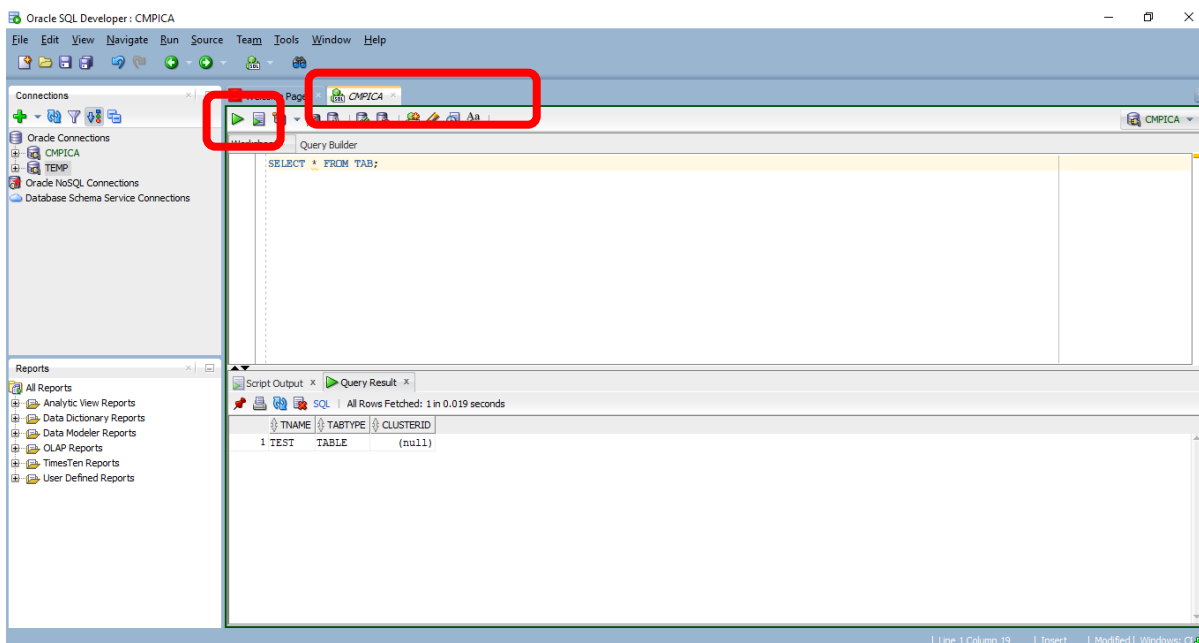


Your newly created connection should get displayed on the left panel.

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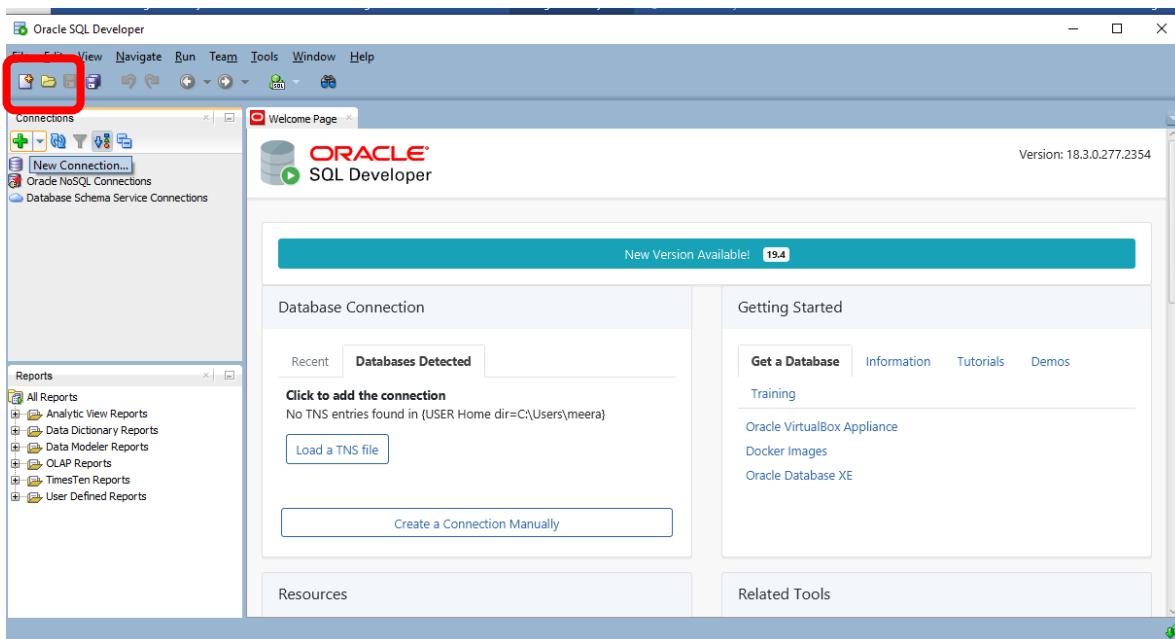
**Step 5** On the right side panel observe the Tab with CMPICA (Connection Name). Write your query and Run your query (execute) using  icon **or** press Ctrl + Enter.



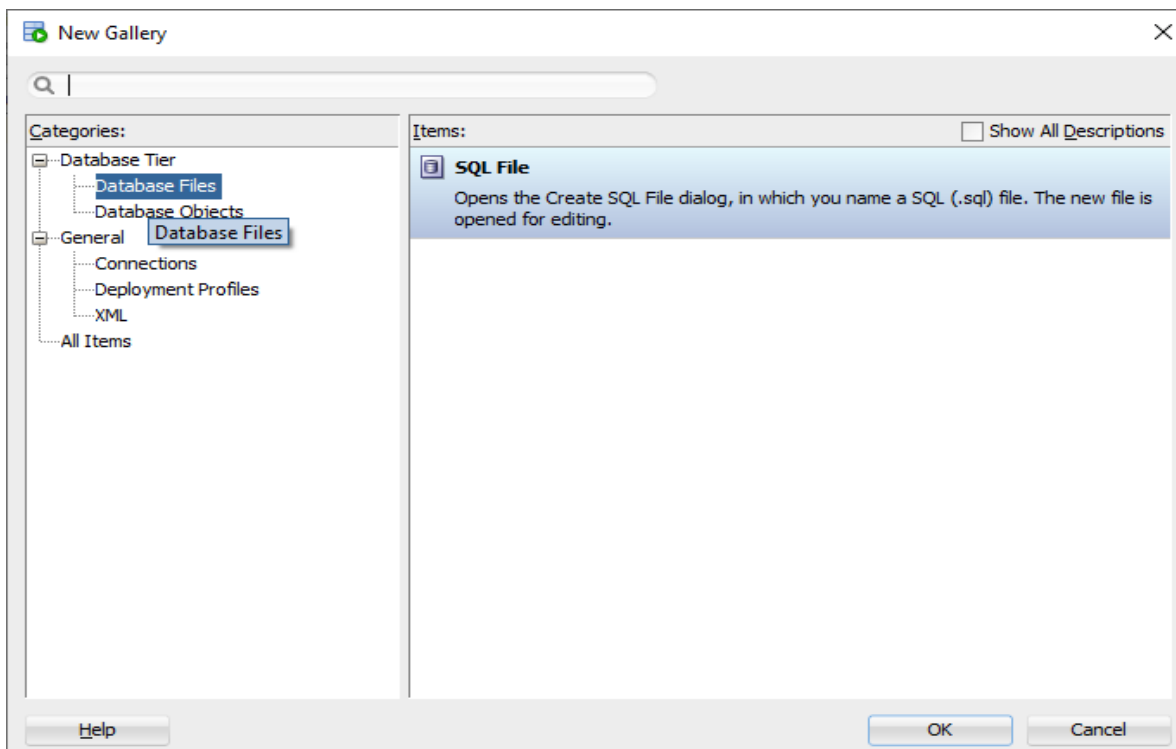
**Step 6** IF you want to create a new SQL file then

In the right panel click on File -> New symbol,  or press Ctrl + N and Then select database -> SQL File and click OK. Please refer below screenshots.

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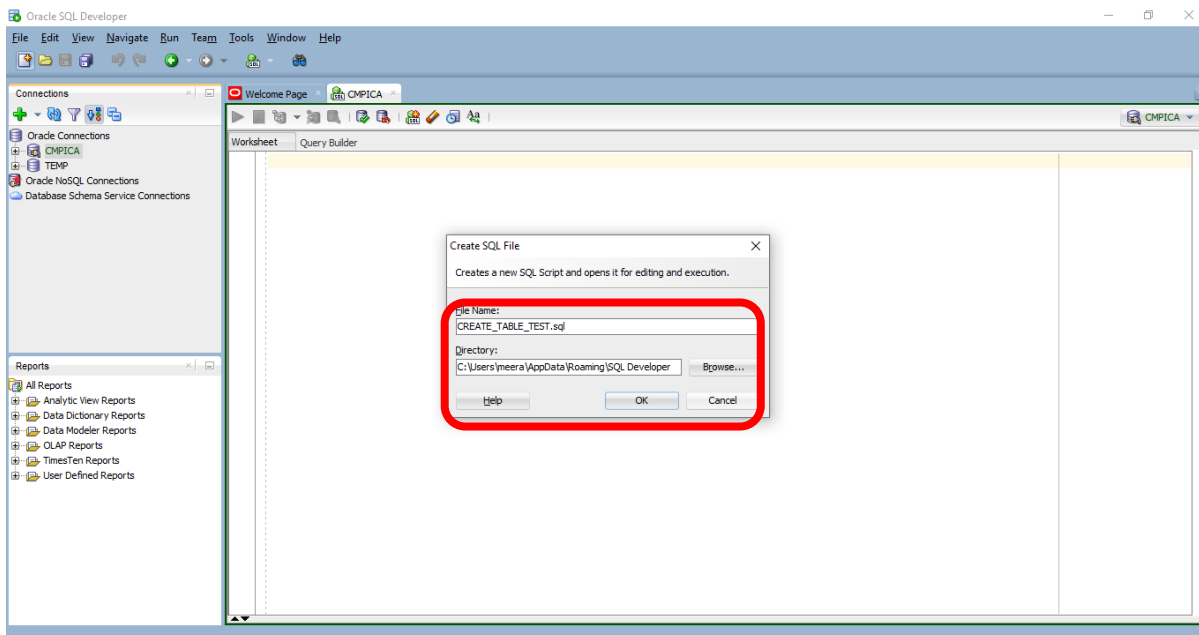


The following screen will appear.



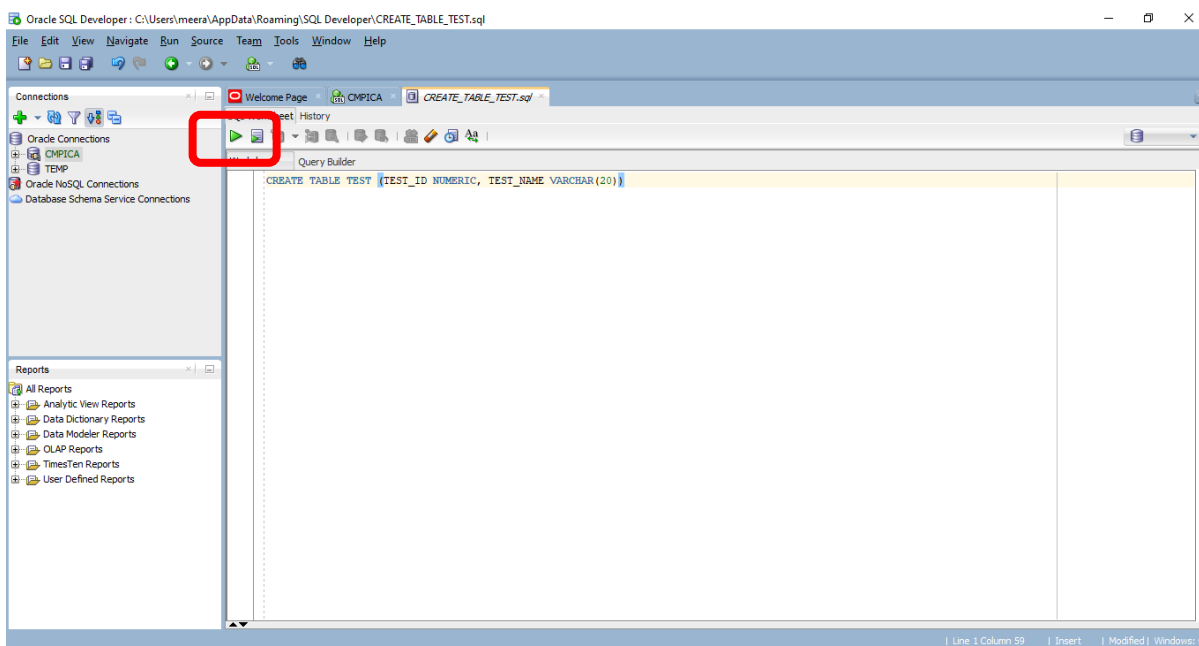
Give proper file name and save to the proper location. [IT IS ADVISED TO SAVE ALL THE SQL IN ONE FOLDER]

## Tool Based Manual || BCA || SEM IV || CA222 : Advanced Database System



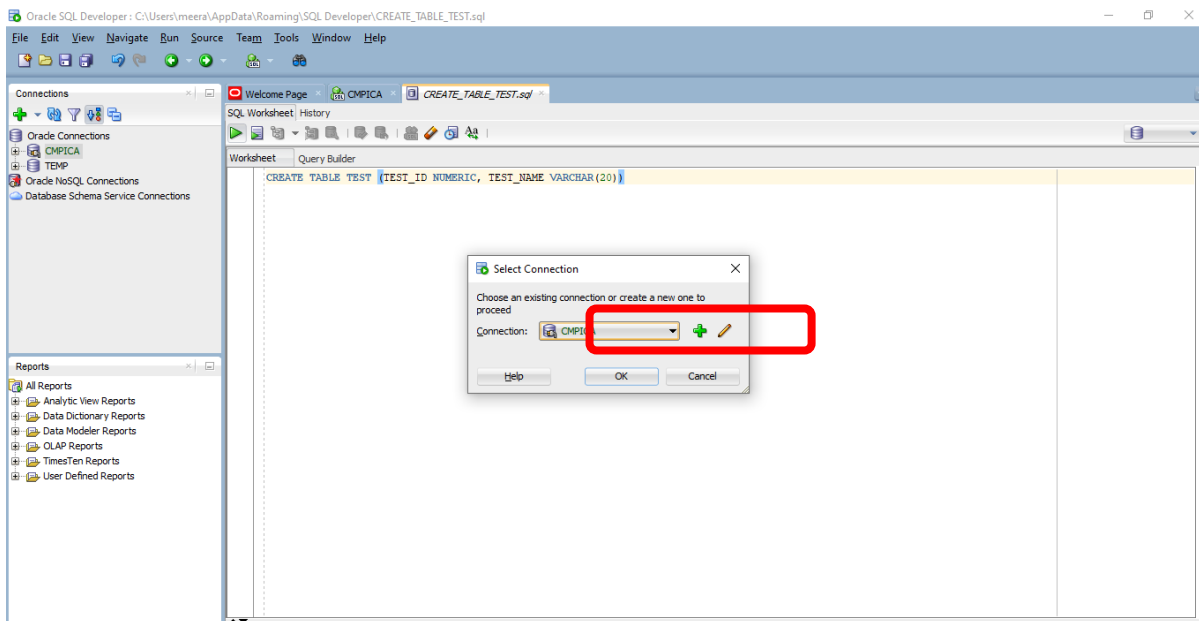
Click OK.

**Step 7** Write your query and Run your query (execute) using  icon **or** press Ctrl + Enter. Select the appropriate Connection in the dialogue box and press OK.

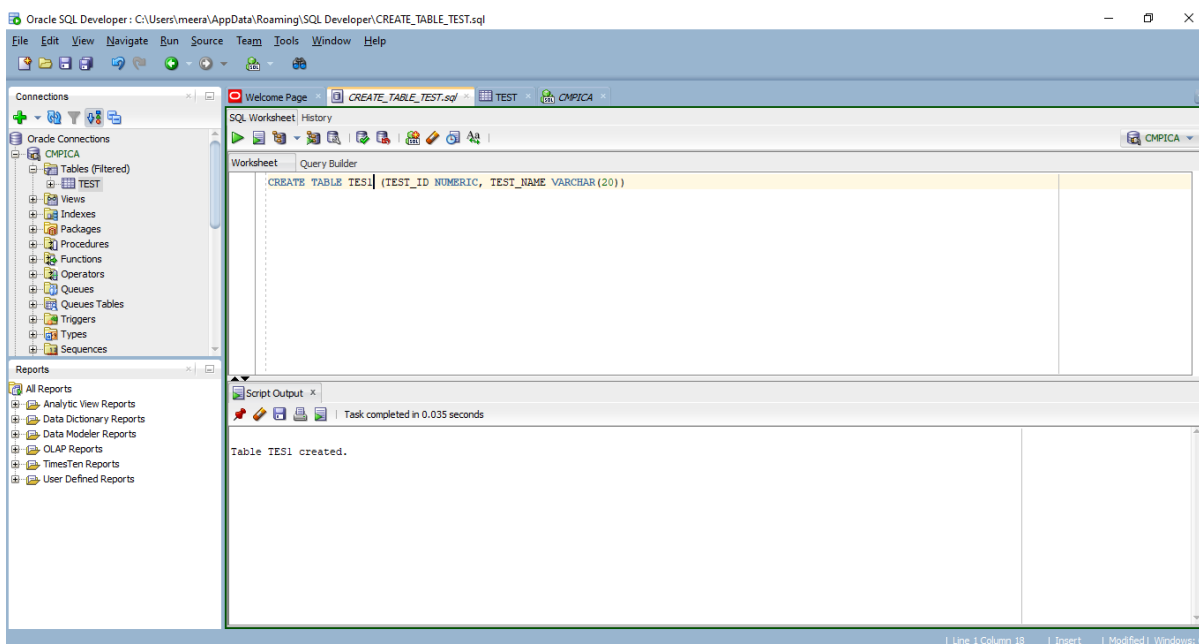


Select the appropriate connection and press OK.

## Tool Based Manual || BCA || SEM IV || CA222 : Advanced Database System



Observe execution result in bottom panel.



**Congratulations!!! Your Oracle SQL Developer Setup is completed successfully.**

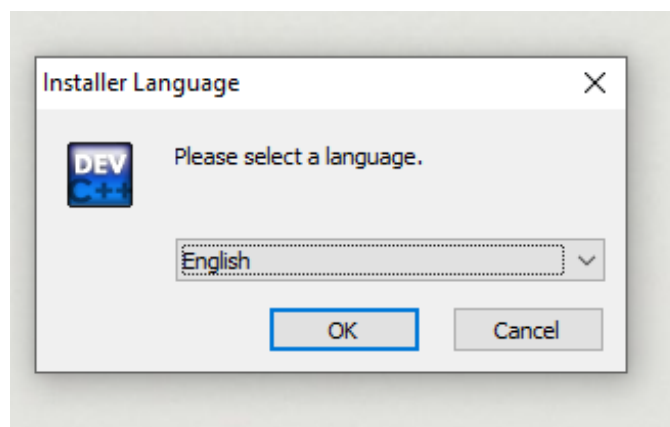
**Happy Database Exploration!!!**



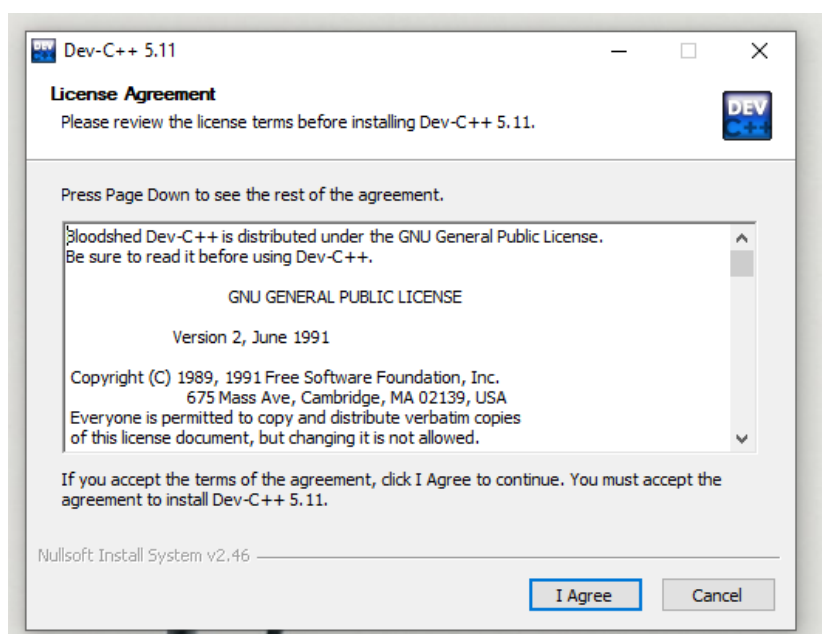
## Installation and Getting Started under Windows OS

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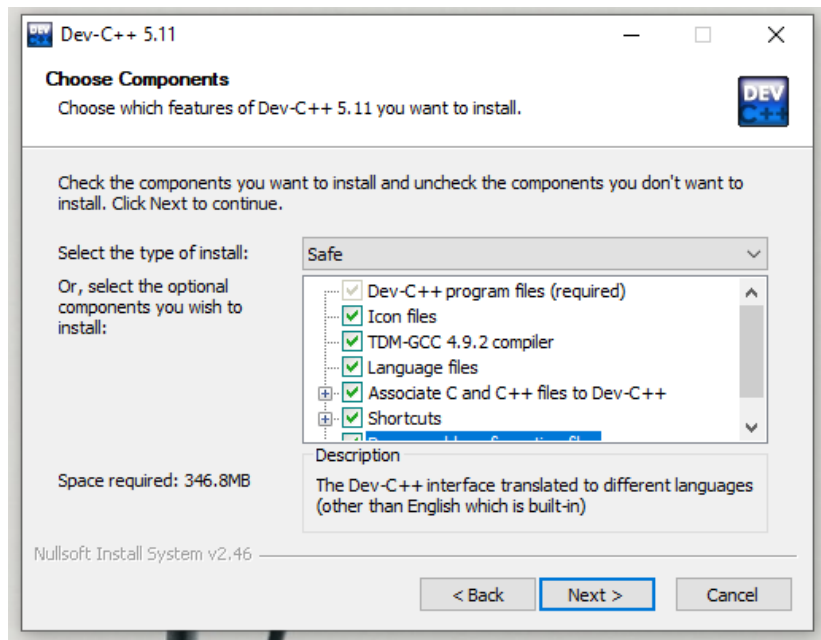


4. Click On Ok.

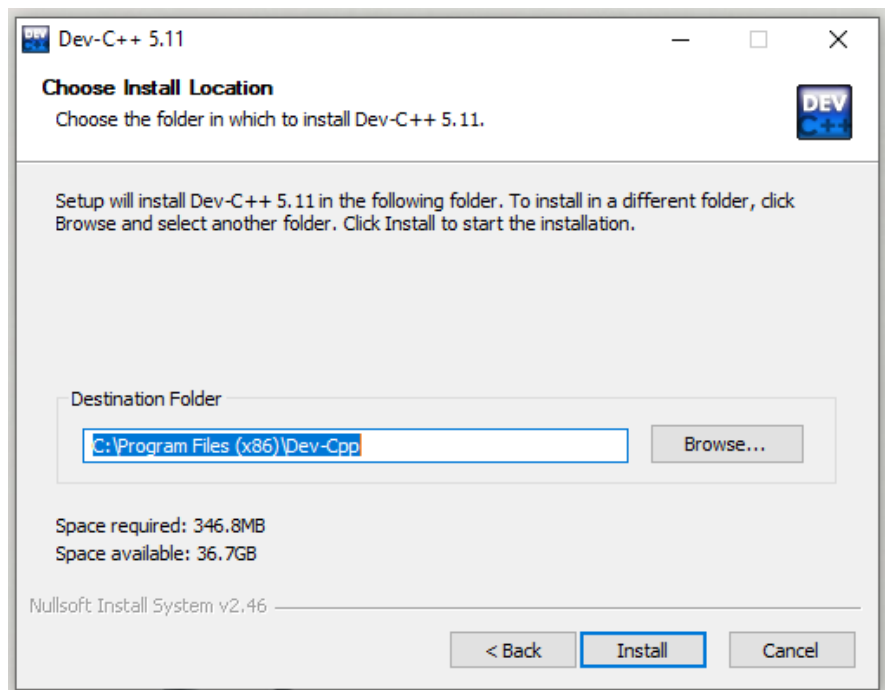


## Tool Based Manual || B.Sc. (IT) || SEM II || CA506 : Data Structure and Algorithms

5. Click On I Agree.

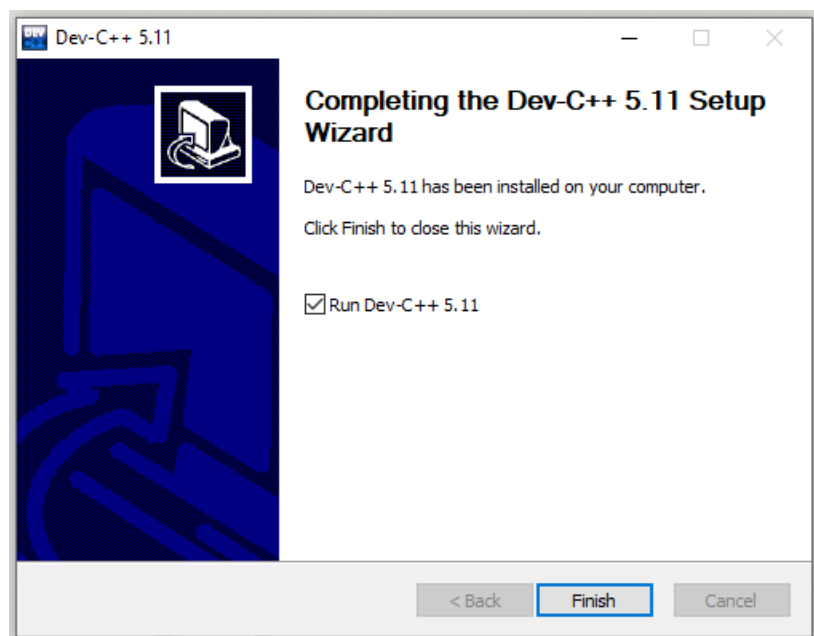


6. Click on Next.

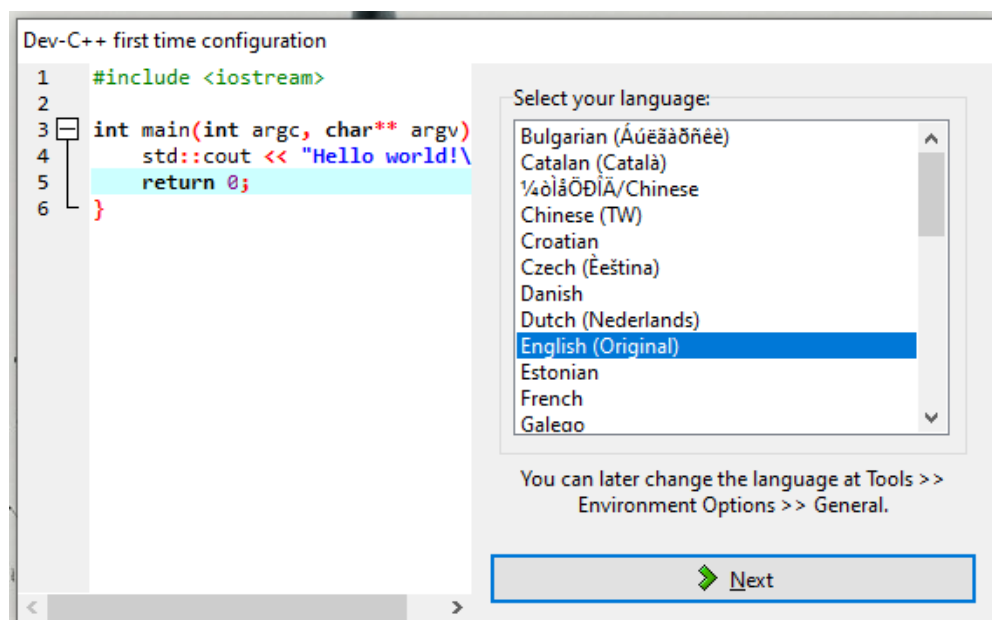


## Tool Based Manual || B.Sc. (IT) || SEM II || CA506 : Data Structure and Algorithms

7. Click on Install.



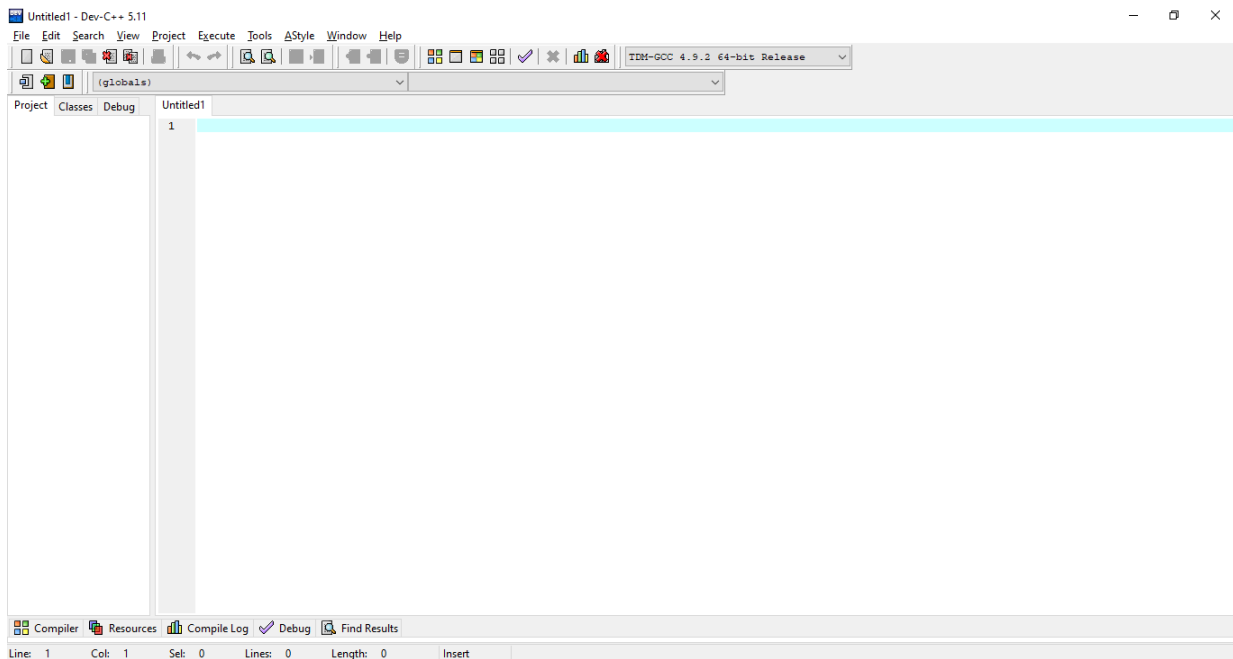
8. Click on finish.



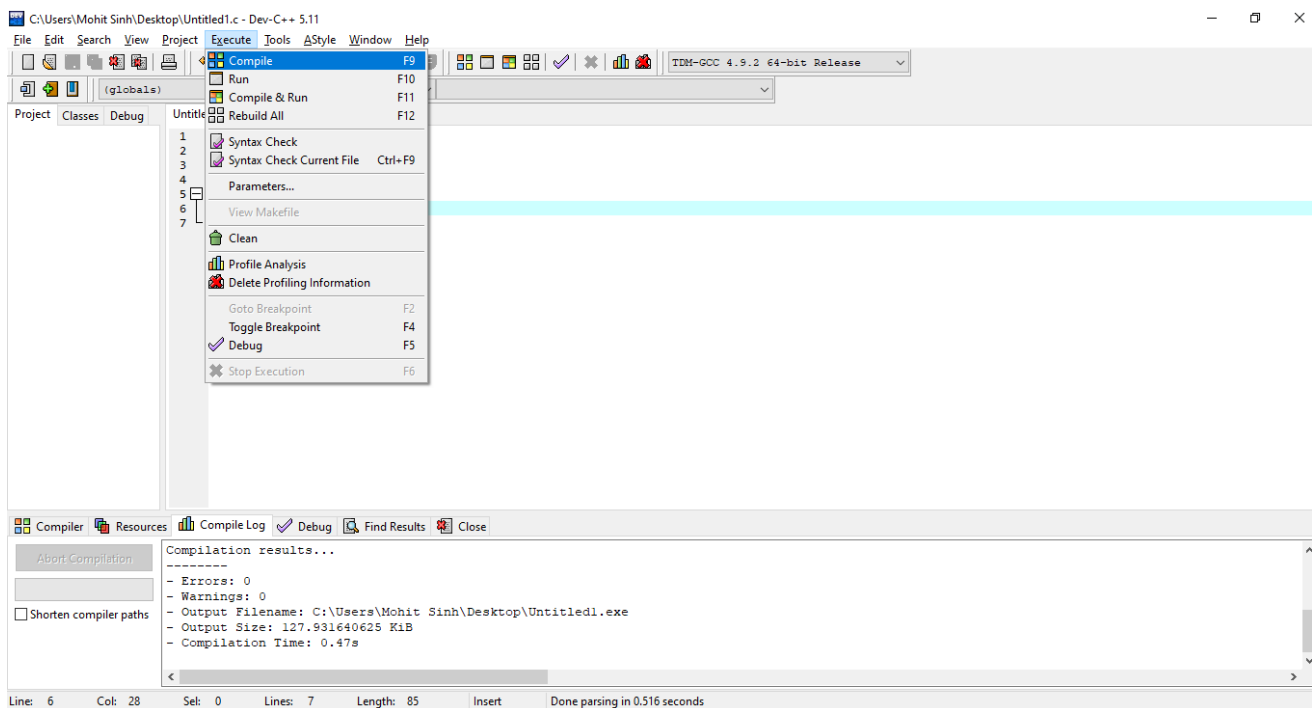
The screenshot shows the Dev-C++ 5.11 application window. The 'File' menu is open, and the 'Project' option is selected, which has opened a submenu. The submenu contains the following items: 'New' (Source File, Ctrl+N), 'Project...' (highlighted), 'Project Template...', and 'Class...'. The main menu also includes 'Open...', 'Save', 'Save As...', 'Save Project As...', 'Save All' (Shift+Ctrl+S), 'Close', 'Close Project', 'Close All' (Shift+Ctrl+W), 'Properties...', 'Import', 'Export', 'Print...', 'Print Setup...', and 'Exit' (Alt+F4). The status bar at the bottom shows 'Compiler', 'Resources', 'Compile Log', 'Debug', and 'Find Results'.

## Tool Based Manual || B.Sc. (IT) || SEM II || CA506 : Data Structure and Algorithms

### 11. IDE of Dev C++.

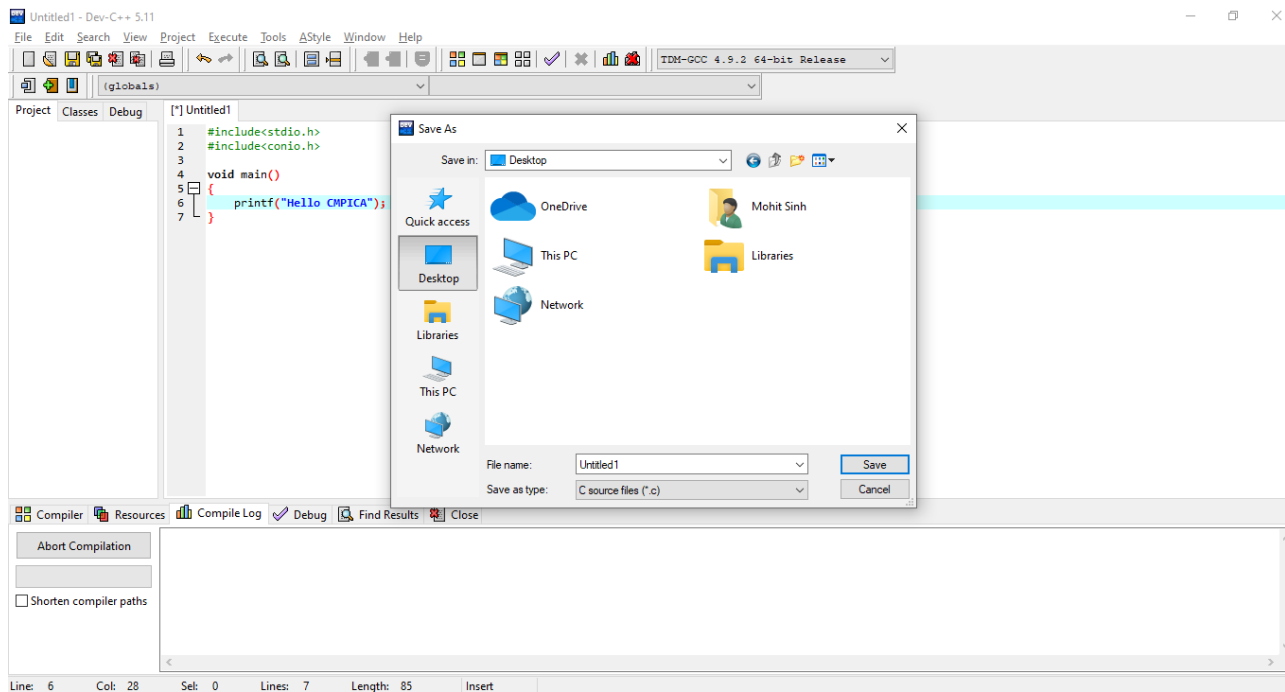


### 12. New file for c programming.

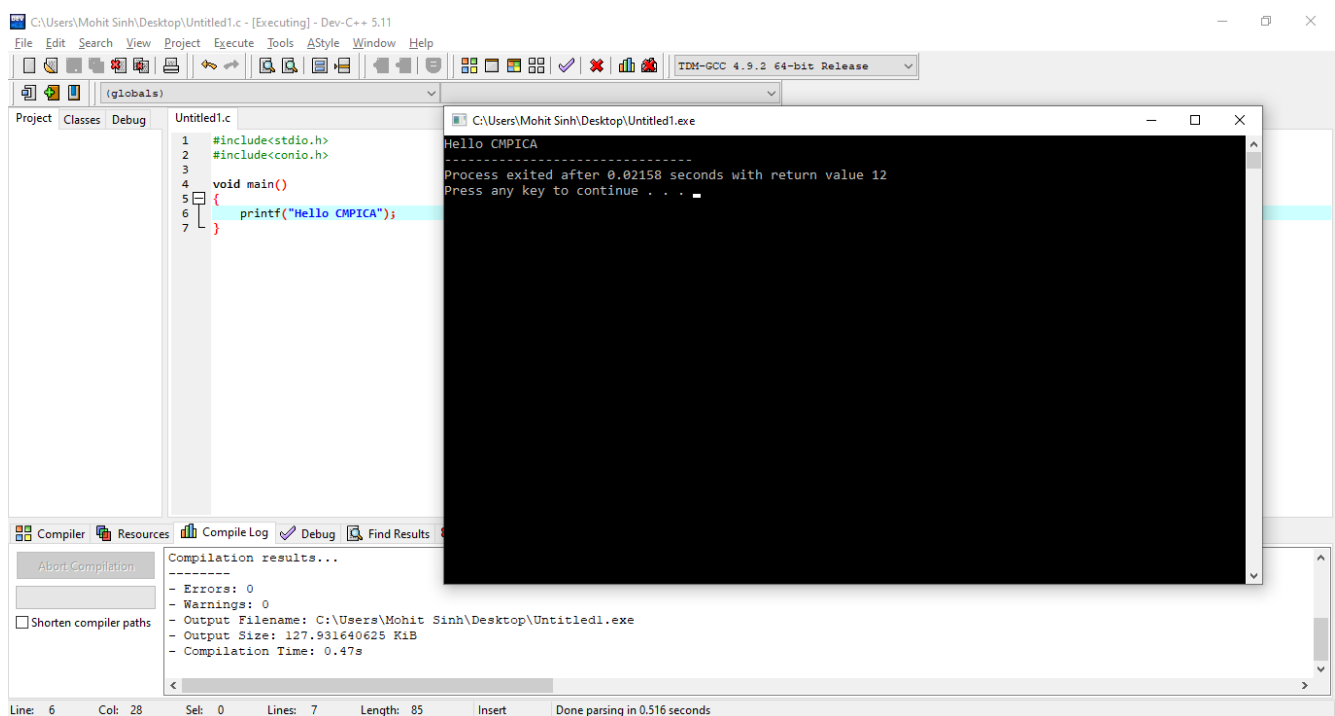


## Tool Based Manual || B.Sc. (IT) || SEM II || CA506 : Data Structure and Algorithms

13. From the execute tab of menu bar you have to select compile option.



14. To run the program, save the file with appropriate name and extension.



The black screen window is the console window that display the output of the program.



## Software Testing Steps of Using Selenium WebDriver with Chrome Browser

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| Software testing. .... | 6 |





## Preview

Google Chrome dominates the browser market with a massive 64% global market share. With a number of useful features, it is one of the most preferred browsers in the world. Given its importance and usage, it becomes crucial to test all websites on Chrome.

A ChromeDriver is a standalone server or a separate executable that is used by Selenium WebDriver to control Chrome. It is impossible to run Selenium test scripts on the Google Chrome browser without ChromeDriver. One can easily initialize the object of ChromeDriver using the following command:

```
WebDriver driver = new ChromeDriver
```



## How to configure ChromeDriver

**Tools required:** ChromeDriver

**Table to Download links for tools:**

|                                                                        |                                                                                                       |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <a href="https://chromedriver.chromium.org/downloads">ChromeDriver</a> | <a href="https://chromedriver.chromium.org/downloads">https://chromedriver.chromium.org/downloads</a> |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|

1. Once the zip file is downloaded for the operating system, unzip it to retrieve the chromedriver.exe executable file. Copy this file to a specific location of your choice.
2. Now copy the path where the ChromeDriver file is saved to set the system properties in environment variables. Follow the steps below to set the path in the environment variables:
  - a. Right-click on **My Computer** and click on **Properties**
  - b. Click on the **Change settings** option and then click on the **Advanced tab**
  - c. Now, from the available options under system variables, select the **Path** option and click on **Edit**
  - d. At the end of the string, enter a semicolon ';' and paste the path of your ChromeDriver file that you copied earlier, and click OK.

Reference: [How to run Selenium tests on Chrome using ChromeDriver | BrowserStack](#)



## Software Testing Using Junit in NetBeans

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## Preview

JUnit is an open-source Unit Testing Framework for JAVA. It is useful for Java Developers to write and run repeatable tests. Erich Gamma and Kent Beck initially develop it. It is an instance of xUnit architecture. As the name implies, it is used for Unit Testing of a small chunk of code.

Developers who are following test-driven methodology must write and execute unit test first before any code.

NetBeans is an open-source integrated development environment (IDE) for developing with Java, PHP, C++, and other programming languages. NetBeans is also referred to as a platform of modular components used for developing Java desktop applications.

## Software testing.

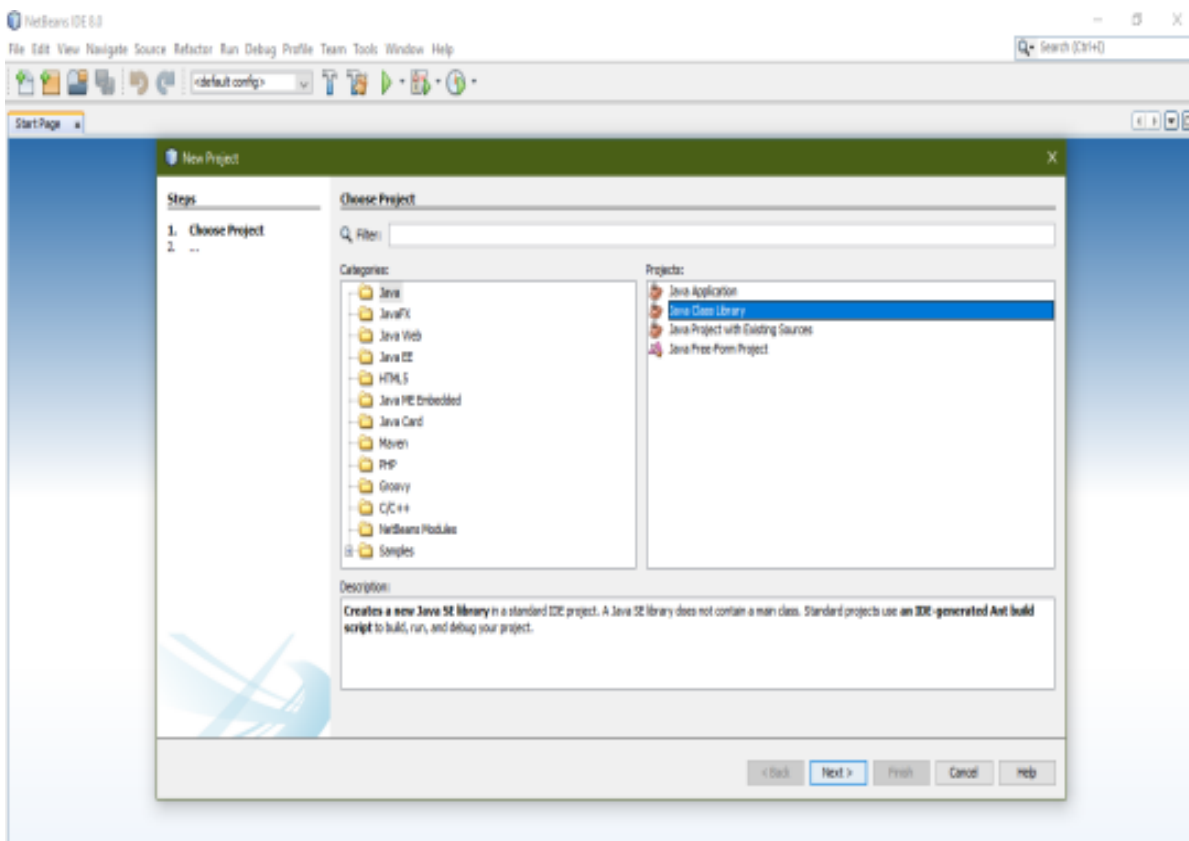
**Tools required:** Junit & NetBeans

**Table to Download links for tools:**

|                                                                               |                                                                                                                                   |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <a href="https://search.maven.org/search?q=g:junit%20AND%20a:junit">JUnit</a> | <a href="https://search.maven.org/search?q=g:junit%20AND%20a:junit">https://search.maven.org/search?q=g:junit%20AND%20a:junit</a> |
| <a href="https://netbeans.apache.org/download/nb126/nb126.html">NetBeans</a>  | <a href="https://netbeans.apache.org/download/nb126/nb126.html">https://netbeans.apache.org/download/nb126/nb126.html</a>         |

1. Open NetBeans IDE
2. Choose Run > Set Main Project in the main menu and select the JUnit-Sample project.
3. Choose Run > Test Project (JUnit-Sample) from the main menu.
4. Choose Window > IDE Tools > Test Results to open the Test Results window.

Reference: [Writing JUnit Tests in NetBeans IDE \(apache.org\)](https://www.apache.org/docs/1.9.0/developing.html)



## Tool Based Manual || MCA || SEM II || CA870 : Software Quality Assurance

